

CBSE Class 12 Physics — Current Electricity

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark Questions (Objective/VSA)

Q1. A steady current of **8 mA** flows through a wire. The number of electrons passing through a cross-section of the wire in **10 s** is: (a) 4.0×10^{16} (b) 5.0×10^{17} (c) 1.6×10^{16} (d) 1.0×10^{17} . [PYQ 2023] Very High

Q2. A potential difference of **200 V** is maintained across a conductor of resistance **100 Ω** . The number of electrons passing through it in **1 s** is: (a) 1.25×10^{19} (b) 2.5×10^{18} (c) 1.25×10^{10} (d) 2.5×10^{16} . [PYQ 2023] High

Q3. A conductor of **10 Ω** is connected across a **6 V** ideal source. The power supplied by the source to the conductor is: (a) **1.8 W** (b) **2.4 W** (c) **3.6 W** (d) **7.2 W**. [PYQ 2023] Very High

Q4. The specific resistance (resistivity) of a thick rod of copper as compared to that of a thin wire of copper is: (a) less (b) more (c) same (d) depends upon the length and area of cross-section of the wire. [PYQ 2022] Very High

Q5. By increasing the temperature, the specific resistance of a conductor and a semiconductor respectively: (a) increases for both (b) decreases for both (c) increases for a conductor and decreases for a semiconductor (d) decreases for a conductor and increases for a semiconductor. [PYQ 2021] Very High

Q6. An electric current is passed through a circuit containing two wires of the same material, connected across the same potential difference. If the lengths and radii of the wires are in the ratio of **3 : 2** and **2 : 3**, then the ratio of the current passing through the wires will be: (a) **2 : 3** (b) **3 : 2** (c) **8 : 27** (d) **27 : 8**. [PYQ 2021] High

Q7. A constant voltage is applied between the two ends of a uniform metallic wire, and heat **H** is developed in it. If another wire of the same material, double the radius and twice the length as compared to the original wire is used, then the heat developed in it will be: (a) **$H/2$** (b) **H** (c) **$2H$** (d) **$4H$** . [PYQ 2021] High

Q8. If the potential difference **V** applied across a conductor is increased to **$2V$** with its temperature kept constant, the drift velocity of the free electrons in the conductor will: (a) remain the same (b) become half of its previous value (c) be double of its initial value (d) become zero. [PYQ 2021] Very High

Q9. A battery is connected to a conductor of non-uniform cross-sectional area. The quantity which remains strictly constant along the length of the conductor is: (a) electric field only (b) drift speed and electric field (c) electric field and current (d) current only. [PYQ 2021] Very High

Q10. A straight line plot showing the terminal potential difference V of a cell as a function of current I drawn from it has intercepts at $V = 6 \text{ V}$ (on the y-axis) and $I = 12 \text{ A}$ (on the x-axis). The internal resistance of the cell would be: (a) 2.8Ω (b) 1.4Ω (c) 0.5Ω (d) Zero. [PYQ 2021] High

Q11. Which of the following has a negative temperature coefficient of resistivity? (a) Metal (b) Semiconductor (c) Metal and semiconductor (d) Metal and alloy. [PYQ 2021] Very High

Q12. Calculate the number of units of electricity used if a bulb of 100 W is kept on for **5 hours**. (a) **1 unit** (b) **0.1 unit** (c) **5 units** (d) **0.5 units**. [PYQ 2020] Medium

Q13. In a current-carrying conductor, the ratio of the electric field and the current density at a point is strictly called: (a) resistivity (b) conductivity (c) resistance (d) mobility. [PYQ 2020] Very High

Q14. The resistance of a metal wire increases with increasing temperature on account of: (a) decrease in free electron density (b) decrease in relaxation time (c) increase in mean free path (d) increase in the mass of the electron. [PYQ 2020] Very High

Q15. The electric power consumed by a 220 V , 100 W bulb when operated at 110 V is: (a) 25 W (b) 30 W (c) 35 W (d) 45 W . [PYQ 2020] High

Q16. A cell of internal resistance r connected across an external resistance R can supply maximum current when: (a) $R = r$ (b) $R > r$ (c) $R = r/2$ (d) $R = 0$. [PYQ 2020] High

Q17. The storage battery of a car has an emf of 12 V . If the internal resistance of the battery is 0.4Ω , the maximum current that can be drawn from the battery is: (a) 15 A (b) 30 A (c) 12 A (d) 20 A . [PYQ 2022] Very High

Q18. We specifically use alloys like manganin for making standard resistors because they have: (a) low temperature coefficient of resistivity and high specific resistance (b) high temperature coefficient and low specific resistance (c) low temperature coefficient and low specific resistance (d) high temperature coefficient and high specific resistance. [PYQ 2021] Very High

Q19. Assertion (A): Kirchhoff's junction rule is based on the conservation of charge. **Reason (R):** A resistor obeys Ohm's law while a diode does not. [PYQ 2022] High

Q20. Assertion (A): The drift velocity of electrons in a metallic wire will decrease, if the temperature of the wire is increased. **Reason (R):** On increasing temperature, the collision frequency of electrons with the lattice ions increases, resulting in a decrease of relaxation time. [Practice] Very High

2-Mark Questions

Q21. Differentiate clearly between electrical resistance and electrical resistivity of a conductor. [PYQ 2020] Very High

Q22. A potential difference V is applied across the ends of a uniform copper wire of length l and diameter D . What is the effect on the drift velocity of electrons if the applied voltage V is exactly halved? [PYQ 2015] Very High

- Q23.** In the context of the same copper wire (length l , diameter D , potential difference V), what will be the effect on the drift velocity if the length l is doubled? [PYQ 2015] Very High
- Q24.** Define the term 'mobility' of charge carriers in a current-carrying conductor. Write its relation in terms of relaxation time. [PYQ 2020] High
- Q25.** Two cells of emfs 1.5 V and 2.0 V having internal resistances $0.2\ \Omega$ and $0.3\ \Omega$ respectively, are connected in parallel. Calculate the equivalent emf of the combination. [PYQ 2016] Very High
- Q26.** The emf of a cell is always strictly greater than its terminal voltage when drawing current. Why? Give the mathematical reason. [PYQ 2022] High
- Q27.** Plot a graph showing the variation of current versus voltage in a semiconductor device like GaAs. Mark the region showing negative resistance. [PYQ 2015] Medium
- Q28.** Under what exact condition will the total current in an external resistance R be the same when connected across a series combination and a parallel combination of n identical cells (each of internal resistance r)? [PYQ 2019] High
- Q29.** A battery of emf 12 V and internal resistance $2\ \Omega$ is connected to a $4\ \Omega$ resistor. Show that a voltmeter when placed across the cell and across the resistor, in turn, gives the exact same reading. [PYQ 2016] High
- Q30.** Why is a voltmeter always placed in parallel and an ammeter always in series in an electric circuit? [PYQ 2016] Medium
- Q31.** Two electric bulbs P and Q have their resistances in the ratio of $1 : 2$. They are connected in series across a battery. Find the ratio of the power dissipation in these bulbs. [PYQ 2022] High
- Q32.** Define the temperature coefficient of resistivity. Write its SI unit. [Practice] Medium

3-Mark Questions

- Q33.** Define the term 'conductivity' of a metallic wire. Using the concept of free electrons in a conductor, derive the expression for the conductivity of a wire in terms of number density (n) and relaxation time (τ). [PYQ 2018] Very High
- Q34.** Using the concept of drift velocity of charge carriers in a conductor, deduce the fundamental relationship between current density (J) and the applied electric field (E). [PYQ 2015] Very High
- Q35.** Show, on a plot, the variation of resistivity of (i) a typical metal conductor, and (ii) a typical semiconductor as a function of temperature. Using the analytical expression for resistivity, explain the respective reasons for their behavior with a rise in temperature. [PYQ 2019] Very High
- Q36.** Deduce the relation between the electric current (I) flowing through a conductor and the drift velocity (v_d) of the free electrons. [PYQ 2015] High

Q37. Two cells of emfs E_1 and E_2 and internal resistances r_1 and r_2 are connected in parallel, with their positive terminals connected to a common point. Obtain the general expressions for the equivalent emf (E_{eq}) and equivalent internal resistance (r_{eq}) of the combination. [PYQ 2023] Very High

Q38. A variable external resistor R is connected across a cell of emf E and internal resistance r . Plot the graph showing the variation of potential drop (V) across R as a function of R . State the value of R at which the current in the circuit is strictly maximum. [PYQ 2020] High

Q39. State Kirchhoff's Junction Rule and Kirchhoff's Loop Rule used in the analysis of electric circuits. Justify the fundamental conservation principle each rule is based upon. [PYQ 2017] Very High

Q40. Derive the strict equation of the balanced state in a Wheatstone bridge using Kirchhoff's laws. [PYQ 2020] Very High

Q41. A 10 V cell of negligible internal resistance is connected in parallel across a battery of emf 200 V and internal resistance $38\ \Omega$. The positive terminals are connected together. Using Kirchhoff's rules, find the value of the current in the circuit. [PYQ 2018] High

Q42. The potential difference applied across a given ohmic resistor is altered so that the heat produced per second increases by a factor of 9 . By what exact factor does the applied potential difference change? [PYQ 2017] Medium

5-Mark Questions (Long Answer/Case-Study)

Q43. (A) Explain the term drift velocity of electrons in a conductor. Hence, derive $v_d = \frac{eE}{m}\tau$. (B) A conducting wire whose cross-sectional area is increasing linearly from its one end to the other is connected across a DC battery of V volts. Which of the following quantities remains strictly constant in the wire, and why? (i) drift speed (ii) current density (iii) electric current (iv) electric field. [PYQ 2017] Very High

Q44. (A) Distinguish clearly between the electromotive force (EMF) and the terminal voltage of a cell having internal resistance r . (B) Draw a plot showing the variation of terminal voltage (V) versus the current (I) drawn from the cell. Explain how one can determine the internal resistance of the cell using this plot. (C) The storage battery of a car has an emf of 12 V . If the internal resistance of the battery is $0.4\ \Omega$, what is the maximum current that can be drawn from the battery? [PYQ 2014] Very High

Q45. (A) Define electrical power. (B) Two non-ideal cells of emfs E_1 and E_2 have internal resistances r_1 and r_2 , respectively. They are connected in series supporting each other across an external resistance R . Deduce the mathematical expression for the equivalent emf and internal resistance of the combination. (C) If the two cells are completely identical, each of emf $E = 5\text{ V}$ and internal resistance $r = 2\ \Omega$, calculate the voltage drop across an external resistance of $R = 10\ \Omega$. [PYQ 2020] High

Q46. Case Study: Temperature Dependence of Resistivity The electrical resistivity of materials depends heavily on their temperature. For pure metals, resistivity increases with an

increase in temperature because the collision frequency of free electrons with the lattice ions increases, which reduces the average relaxation time. Conversely, for semiconductors, the resistivity drops exponentially as temperature rises due to the breaking of covalent bonds, exponentially increasing the number density of charge carriers. Alloys like nichrome and manganin show very weak temperature dependence. (a) Write the empirical formula showing the variation of the resistivity (ρ) of a metal with a small change in temperature (ΔT). (b) Why are specific alloys like constantan or manganin chosen to construct standard wire-bound resistors? (c) A silver wire has an electrical resistance of $2.1\ \Omega$ at 27.5°C , and a resistance of $2.7\ \Omega$ at 100°C . Determine the temperature coefficient of resistivity (α) of silver. [Practice] Very High

Q47. Case Study: Internal Resistance of a Cell When a cell drives a steady current through an external circuit, the charge carriers have to pass through the electrolyte within the cell itself. The electrolyte offers a natural resistance to the flow of current. This opposition is called the internal resistance (r) of the cell. Due to this internal resistance, the terminal voltage (V) of a discharging cell is always less than its electromotive force (E). (a) Define the electromotive force (EMF) of a cell. (b) Write the mathematical relationship between terminal voltage V , EMF E , current I , and internal resistance r for a cell that is discharging. (c) A cell of EMF $12\ \text{V}$ and internal resistance $2\ \Omega$ is connected across a $4\ \Omega$ external resistor. Calculate the terminal voltage of the cell. [Practice] High

Q48. (A) What is a Wheatstone bridge? Draw its standard circuit diagram. (B) State the condition for the bridge to be balanced. (C) In a Wheatstone bridge setup, the four resistive arms have values $10\ \Omega$, $20\ \Omega$, $30\ \Omega$, and $60\ \Omega$ in cyclic order. Is this specific bridge balanced? Show mathematically. [PYQ 2022] Very High

Q49. (A) Deduce Ohm's law macroscopically from the microscopic concept of drift velocity and relaxation time. (B) A straight horizontal conducting wire of resistance R is cut into three equal parts. If these three parts are now tied together in parallel to form a new bundled wire, what will be the resistance of this bundled wire in terms of R ? [Practice] High

Q50. (A) Define the term mobility of charge carriers. State its SI unit. (B) Estimate the average drift speed of conduction electrons in a copper wire of cross-sectional area $1.0 \times 10^{-7}\ \text{m}^2$ carrying a steady current of $1.5\ \text{A}$. Assume the number density of conduction electrons in copper to be $9 \times 10^{28}\ \text{m}^{-3}$. (Take $e = 1.6 \times 10^{-19}\ \text{C}$) [PYQ 2014] Very High

Answer Key

Q1: (b) 5.0×10^{17} . ($q = It = 8 \times 10^{-3} \times 10 = 0.08\ \text{C}$).

$n = q/e = 0.08/(1.6 \times 10^{-19}) = 5 \times 10^{17}$. **Q2:** (a) 1.25×10^{19} . ($I = V/R = 2\ \text{A}$).

$n = I/e = 2/(1.6 \times 10^{-19}) = 1.25 \times 10^{19}$. **Q3:** (c) $3.6\ \text{W}$. ($P = V^2/R = 36/10 = 3.6\ \text{W}$).

Q4: (c) same. (Specific resistance / resistivity depends only on the nature of the material and temperature, not on dimensions). **Q5:** (c) increases for a conductor and decreases for a semiconductor. **Q6:** (c) $8 : 27$. ($R \propto l/r^2$. Ratio of

$R = (3/2) \times (3/2)^2 = 27/8$. Current $I \propto 1/R$ since V is same. $I_1/I_2 = 8/27$. **Q7:** (c) $2H$.
 $(R' = \rho(2l)/\pi(2r)^2 = R/2$. $H = V^2t/R$. New $H' = V^2t/(R/2) = 2H$). **Q8:** (c) be double
 of its initial value. ($v_d = \frac{eV\tau}{ml}$. Since V is doubled, v_d is doubled). **Q9:** (d) current only.
 (Current is conserved at any cross-section. Drift speed, current density, and E-field vary with
 area). **Q10:** (c) 0.5Ω . ($V = E - Ir$. At $I = 0$, $V = E = 6 \text{ V}$. At $V = 0$, $I = 12 \text{ A}$.
 $0 = 6 - 12r \Rightarrow r = 0.5 \Omega$). **Q11:** (b) Semiconductor. (Resistivity drops as temperature
 increases). **Q12:** (d) 0.5 units . (Energy (kWh) = $\frac{100 \text{ W} \times 5 \text{ h}}{1000} = 0.5 \text{ kWh}$). **Q13:** (a) resistivity.
 $(J = \sigma E \Rightarrow E/J = 1/\sigma = \rho)$. **Q14:** (b) decrease in relaxation time. (Due to increased
 thermal collisions). **Q15:** (a) 25 W . ($R = V^2/P = 220^2/100 = 484 \Omega$. At 110 V ,
 $P' = 110^2/484 = 25 \text{ W}$). **Q16:** (d) $R = 0$. ($I = E/(R + r)$. I is max when external
 resistance is zero, giving short circuit current E/r). **Q17:** (b) 30 A . (
 $I_{max} = E/r = 12/0.4 = 30 \text{ A}$). **Q18:** (a) low temperature coefficient of resistivity and high
 specific resistance. **Q19:** (b) Both A and R are true but R is not the correct explanation of A.
Q20: (a) Both A and R are true and R is the correct explanation of A. **Q21:** Resistance is the
 opposition to current flow ($R = \rho l/A$), depending on dimensions. Resistivity (ρ) is the
 inherent property of a material, independent of shape or size. **Q22:** Drift velocity is halved. (
 $v_d \propto V$). **Q23:** Drift velocity is halved. ($v_d \propto 1/l$). **Q24:** Mobility (μ) is the magnitude of drift
 velocity per unit electric field. $\mu = v_d/E = e\tau/m$. **Q25:**
 $E_{eq} = \frac{E_1r_2 + E_2r_1}{r_1 + r_2} = \frac{(1.5)(0.3) + (2.0)(0.2)}{0.2 + 0.3} = \frac{0.45 + 0.40}{0.5} = 1.7 \text{ V}$. **Q26:** During discharge,
 $V = E - Ir$. Since Ir is positive, $V < E$. The difference is the potential drop across the
 internal resistance. **Q27:** Draw the standard GaAs V-I curve showing an initial linear rise,
 reaching a peak, then dropping (negative slope region), and finally rising again. Mark the
 downward sloping region as negative resistance. **Q28:** The current is the same when external
 resistance equals the internal resistance of one cell, i.e., $R = r$. **Q29:** Across resistor:
 $V = IR = (\frac{12}{4+2}) \times 4 = 2 \times 4 = 8 \text{ V}$. Across cell: $V = E - Ir = 12 - (2)(2) = 8 \text{ V}$.
 They are identical. **Q30:** Voltmeter has very high resistance; placed in parallel, it draws
 negligible current. Ammeter has very low resistance; placed in series, it doesn't drop
 significant voltage. **Q31:** In series, current I is constant. Power $P = I^2R \Rightarrow P \propto R$. Ratio
 $P_P/P_Q = R_P/R_Q = 1 : 2$. **Q32:** It is the fractional change in resistivity per degree rise in
 temperature: $\alpha = \frac{1}{\rho_0} \frac{d\rho}{dT}$. SI Unit: K^{-1} or $^\circ\text{C}^{-1}$. **Q33:** $\sigma = 1/\rho$. Drift velocity $v_d = \frac{eE}{m}\tau$.
 Current $I = n e A v_d = n e A (\frac{eE}{m}\tau)$. $I/A = n e^2 E \tau / m \Rightarrow J = \sigma E \Rightarrow \sigma = \frac{n e^2 \tau}{m}$. **Q34:**
 Start with $I = n e A v_d$. Current density $J = I/A = n e v_d$. Put $v_d = \frac{eE}{m}\tau$. Thus
 $J = n e (\frac{eE}{m}\tau) = (\frac{n e^2 \tau}{m}) E$. Since $\sigma = \frac{n e^2 \tau}{m}$, $J = \sigma E$. **Q35:** Metal: Graph curves upward
 (increases). $\rho = m/(n e^2 \tau)$. τ decreases with temp, so ρ increases. Semiconductor: Graph
 curves exponentially downward. n increases exponentially with temp, dominating the τ
 decrease, so ρ drops. **Q36:** Let volume of conductor be Al . Total free electrons = nAl . Total
 charge $q = n e Al$. Time taken to cross l is $t = l/v_d$. Current
 $I = q/t = n e Al / (l/v_d) = n e A v_d$. **Q37:** Using Kirchhoff's rules on parallel branches:
 $I = I_1 + I_2 = \frac{E_1 - V}{r_1} + \frac{E_2 - V}{r_2}$. Rearranging for V : $V = \frac{E_1/r_1 + E_2/r_2}{1/r_1 + 1/r_2} - I \frac{1}{1/r_1 + 1/r_2}$.
 Comparing to $V = E_{eq} - I r_{eq}$ gives the standard equivalent formulas. **Q38:** Graph is an
 asymptotic curve starting from origin ($V = 0$ at $R = 0$) and saturating at $V = E$ as $R \rightarrow \infty$

. Current is maximum when $R = 0$ (short circuit). **Q39:** Junction Rule ($\Sigma I = 0$): Based on the conservation of charge (charges do not accumulate at a node). Loop Rule ($\Sigma V = 0$): Based on the conservation of energy (work done in a closed path in a conservative field is zero).

Q40: Draw the Wheatstone circuit. Apply Kirchhoff's loop laws for the two inner loops.

Impose the balance condition $I_g = 0$ (current through galvanometer is zero). Equate the voltage drops: $I_1 P = I_2 R$ and $I_1 Q = I_2 S$. Dividing gives $P/Q = R/S$. **Q41:** Let current I flow from 200V battery into the 10V battery. Equation:

$200 - I(38) - 10 = 0 \implies 38I = 190 \implies I = 5 \text{ A}$. **Q42:** Heat $H \propto V^2$. If H increases by 9 times, V^2 increases by 9 times. Therefore, V must increase by a factor of $\sqrt{9} = 3$. **Q43:**

(A) The average velocity with which free electrons get drifted towards the positive end of the conductor. $v_d = a\tau = (eE/m)\tau$. (B) Only electric current (iii) remains constant due to conservation of charge. Drift speed, J , and E decrease as Area increases ($v_d \propto 1/A$, $J \propto 1/A$, $E \propto 1/A$). **Q44:** (A) EMF is max potential difference in open circuit. Terminal V is potential difference in closed circuit. (B) Graph is a straight line with negative slope. $V = E - Ir$. Y-intercept is E , negative slope magnitude is r . (C) $I_{max} = E/r = 12/0.4 = 30 \text{ A}$. **Q45:** (A)

Rate at which electrical energy is consumed. (B) Series: $E_{eq} = E_1 + E_2$, $r_{eq} = r_1 + r_2$. (C) $E_{eq} = 5 + 5 = 10 \text{ V}$, $r_{eq} = 2 + 2 = 4 \Omega$. Current $I = 10/(10 + 4) = 10/14 = 5/7 \text{ A}$. Voltage drop $V = IR = (5/7) \times 10 = 50/7 \approx 7.14 \text{ V}$. **Q46:** (a) $\rho_T = \rho_0[1 + \alpha(T - T_0)]$. (b) They have a very low temperature coefficient of resistivity (α), meaning their resistance remains highly stable over a wide temperature range. (c)

$\alpha = \frac{R_2 - R_1}{R_1(T_2 - T_1)} = \frac{2.7 - 2.1}{2.1(100 - 27.5)} = \frac{0.6}{2.1 \times 72.5} \approx 0.00394 \text{ } ^\circ\text{C}^{-1}$. **Q47:** (a) The potential difference across the cell terminals when no current is drawn. (b) $V = E - Ir$. (c) Current $I = 12/(4 + 2) = 2 \text{ A}$. Terminal voltage $V = IR = 2 \times 4 = 8 \text{ V}$. **Q48:** (A) A specific arrangement of four resistors used to accurately measure an unknown resistance. (B)

$P/Q = R/S$ (No current through the central galvanometer). (C) Ratio of first two: $10/20 = 1/2$. Ratio of next two: $30/60 = 1/2$. Since $1/2 = 1/2$, the bridge is perfectly balanced. **Q49:** (A) $I = neAv_d = neA(eE/m)\tau = (ne^2\tau/m)A(V/l)$. Rearranging

$V/I = ml/(ne^2\tau A) = \rho l/A = R$. Thus $V = IR$. (B) A wire cut into 3 parts: each part has resistance $R/3$. If connected in parallel: $1/R_{new} = 3/(R/3) = 9/R \implies R_{new} = R/9$.

Q50: (A) Mobility is drift velocity per unit electric field. Unit: m^2/Vs . (B)

$I = neAv_d \implies v_d = \frac{I}{neA} = \frac{1.5}{9 \times 10^{28} \times 1.6 \times 10^{-19} \times 1.0 \times 10^{-7}} = 1.04 \times 10^{-3} \text{ m/s}$.